Omni Update 11/20

Key Considerations for Designing a Shared Inventory Model

1. Industry learnings and proven operating models
2. Channel ownership, accountability & operating guardrails
3. Overall margin maximization, not channel wins
4. Protection rules to prevent over-allocation & hoarding
5. Inventory positioning - Ensuring the right place at the right time
6. Smart fulfillment logic with multi-node routing
7. Defined returns & reverse logistics pathways
8. High data accuracy and governance

Shared Inventory Models

Model	Definition	Red Polo Example (100 Units)	Who Uses It	Implementation Details
1. Fully Pooled	One physical shared pool at the DC. All channels pull from the same bucket.(first come, first serve).	Wholesale sells 90 first, gets 90.	Small DTC brands, early startups	
2. Inventory Protection (SUP)	Shared pool at DC, BUT each channel has protected minimums (Protection Objects/SUP)	FP 40, WH 30, OP 10 protected; 20 shared.	Adidas	Uses SAP SUP to create channel-specific protection buckets; protected stock adjusts by time window; unprotected stock shared via rules. Pros - Margin protection Cons: Requires periodic rebalancing
3. Rule-Based Access (Dynamic Reallocation)	Virtual pools that rebalance dynamically based on sell-through + time windows.	FP 40, WH 40, OP 20; FP can get more basis need	Nike	Uses Celect predictive analytics + RFID to track inventory in real time; dynamic reallocation based on hyper-local demand; virtual pools rebalance automatically; Pros: Demand driven, flexible Cons: Can ruin margins
4. Actively Managed	Inventory is segmented by channel within the DC. Planners manually decide to move inventory between segments.	Wholesale moves slow; planners move 20 units to FP DTC.		Planner-driven weekly reallocation; manual transfers prioritized based on ROI
5. Fully Segmented	Each channel has completely separate inventory in different physical DCs. Zero sharing.	FP 40, WH 40, OP 20 fixed.		

Nordstrom - Service Centric Strategy

Introduction: Nordstrom implemented inventory pooling, to integrate online and offline stock into one visible network. This allowed any store to fulfill any offline order, turning 115+ retail locations into distribution centers.

Why: So their e-commerce warehouses and physical stores operate separately. If the website warehouse was out of stock then the customer would be turned away, even if the item was left unsold at a physical store. As the cost for this would be eventually a marked down item at the physical store, eroding margins, while the online sale was lost entirely.

Nordstrom - Service Centric Strategy

Primary Goal: Nordstrom’s strategy is fundamentally about being driven by satisfying customers above all.

Inventory Pooling: Integrating inventory ensures product availability across all channels.

Strategic Trade-Off: Nordstrom accepts higher split shipment costs to guarantee the customer gets the product they want.

Nordstrom's Shared Inventory Model

Type: Rules based access - Model 3

Mechanism: Nordstrom uses a dynamic virtual-pool model where real-time RFID + OMS rules decide the best store or DC to fulfill each order.

Access: Any customer order can be served from any location that has the item.

Logic: Uses business rules to decide the best fulfilment node, but availability is universal.

Fulfillment

- ★ **Priority 1 (Distribution Center):** The system first checks the E-commerce distribution center.
- ★ **Priority 2 (Store Fulfillment):** If the distribution center is out of stock, the order isn't cancelled. Instead, the algorithm re-routes the order to a specific retail store that has the item in stock.
- ★ **Logic Factors:**
  - **Proximity:** Which store is closest to the customer?
  - **Markdown Liability:** Is this item about to be marked down in that specific store?
  - **Inventory Depth:** Does the store have 10 items or just 1?

Industry Research - Customer Experience (CX)

"[Effective] Omnichannel management is "the synergetic management of the numerous available channels and customer touchpoints, in such a way that the customer experience across channels and the performance over channels are optimized" (2)

Key Considerations:

- Retail stores as a 'Hub' (1)
  - BOPS for low fulfillment cost, STS for higher
- Online markets have better margins for niche products, while offline markets serve generics (1)
  - Effective redirect systems for greatest profit
  - Customers have preference of certain channels for certain products - correct alignment can drive revenues up to 40% (2)
- Pool data for customer research (1)

Costs and Risks:

- Inconsistent pricing, quality, or options between channels can cause a negative impact on user experience (2)
- Companies with strong offline performance lower profits and increase risk on online channels in an OC environment (2)

What we can take from Leaders:

A meta-analysis of companies with omni-channel distribution was conducted, identifying companies that had seen the most success with their implementation 'leaders' and those who had struggled to transition or seen little benefit 'non-leaders' (3)

Category	Description	% Leaders w/ Capability	% Non-Leader with Capability	Impact for leaders
Post-Order Modifications	Adding, removing, or otherwise changing products after an order has been placed	50%	13%	**24% higher customer satisfaction**
Unified Customer Support	Consistent experiences and seamless integrations across channels	90%	40%	**0.5x support escalations**
Intelligent Cart	Cart remains constant among channels	70%	31%	**20% lower cart abandonment**

Industry Research - Logistics Optimization

What Does Literature Support?

The benefits of a pooled inventory are determined by the size of channels (1)

Ex. If in-person is 95% of orders, no benefit to any channels from pooling

True pooling - using store inventory to fulfill all online orders - is generally ineffective (1)

Reduced store inventories and higher average holding costs

Dynamically deciding the responsible channel brings greatest economic advantage (1, 4)

Up to 31% lower fulfillment costs (4)

Emerging Fields - Stores as Distribution Centers

OC retailers can take advantage of the resources already in their in-person stores to manage some/all online order fulfillment in two ways: (2)

1. STS (Ship to Store) - Distribution center ships to store:
   a. Having the store as an end point now makes all brick and mortar locations into mini distribution centers that act as a halfway point between larger centers and customers
2. BOPS (Buy Online, Pay in-Store) - Store fulfills online orders with their own stock:
   a. Simplifies demand forecasting and makes use of existing supply chains for distributions
   b. Can cause strain on the store's inventory and be ineffective if store cannot handle added volume

Key Considerations: (1)

These models assume stores are already effective at handling their own replenishment

Locations that already move small orders of inventory frequently perform the best.

Larger and slower moving product can "clog up" the system

Returns (2)

- Returns offer a key differentiation point among OC competitors
- Companies in Japan with cross-channel return capabilities saw greatest profit

Additional Considerations (Future Research) (2,3)

Demand forecasting and allocation is just the first step in building an omnichannel system

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Additional considerations need to be taken to enable each channel to properly fulfill the orders they have been assigned

Industry Research - Model Best Practices

What is the Best Model?

Dynamically forecasting demand and assigning fulfillment responsibility, allowing channels to dip into stock when needed (1)

Needs for Implementing

- Dedicated warehouses for online channels

  Why? When online volume is large, stores cannot effectively manage online distributions while upkeep daily tasks
- Real time data analytics and

  Why? There is no universal answer for a store or distribution center being the best option for online fulfillment. Real time analytics need to be in place to assign incoming orders to whichever channel can fill the demand at lower costs
- Enhanced communications between channels

  Why? Dynamic distribution is not possible without state of the art communications between channels
- DOM (Distributed Order Management)

  Why? The most successful firms integrate a customer database that spans across channels. This gives the most accurate demand forecast by location, giving a much clearer answer to where/who should the order be fulfilled

Implementation (2,

Fully realizing a dynamic access system is costly and takes time. Research found that these systems are often best implemented organically, by beginning with simple protections (model 2). Once an effective protections systems is in place, more complex supply chain analysis focused on reducing six key factors:

1. *Lead times*
2. *Transport costs*
3. *Handling costs*
4. *Fixed operations costs*
5. *Holding costs*
6. *Backorder costs*

Literature supports the following process for fully realizing the most effective OC model:

- Begin by implementing a pooled inventory with protections
- Implement data analytics and a DOM to analyze the 6 factors
- Over time, use the data to begin to taper off of a strict reservation system and move into dynamic fulfillment by minimizing the 6 factors based on collected data

Overall Margin Maximization, Not Channel Wins

Protection Limits

Full Price DTC - Max 45%
Highest margin deserves largest protection, but capped to ensure shared pool exists

Wholesale - Max 35%
Contract obligations require protection, but limited to prevent margin erosion

Off Price DTC - Residual
Lower margin channel with no contractual obligation

Shared Pool Priority

Full Price DTC - 1st Priority
Highest margin

Off Price DTC - 2nd Priority

Off Price DTC - 3rd Priority

- These limits help us prevent wholesale from protecting 60% even if they want to, while guaranteeing FP DTC can protect up to 45%.
- Prioritisation of channels is inbuilt in protection limits
- Shared pool priority is used in cases of overlapping orders across channels with limited available inventory

Inventory Positioning & Shared Pool Access

Key Decisions: Storing the inventory at the right place and the right time. Should channels consume their protected stock first, or should the system dynamically route from the shared?

Store Inventory for Omni-Channel

- Stores now fulfill multiple demand sources: walk-in, BOPIS, ship-from-store, ECOM backup

- Require buffer stock beyond traditional selling floor allocation

When Does a Channel Access Shared Pool?

Protection-First Approach

- Access shared pool only after protected stock consumed
- Ensures accountability but may slow fulfillment

Optimal Fulfilment Approach

- System evaluates: protection status, delivery speed, cost, margin
- Access shared pool if faster/cheaper, even with protection remaining
- Unused protection released monthly to shared pool

Channel/Region Accountability Mechanisms

Accountability Approach: Dynamic SUP (Supplier Protection)

- Each channel receives a protected allocation based on a forecast.
- Variable protection levels are set based on product type. Channels are accountable for consuming their protected allocation and for forecast accuracy.
- Accountability metric: **SUP utilization rate** (target >85%)
- Consequence of low utilization: Protection quantities reduced in following periods; unused inventory releases to shared pool.

Regional Accountability:

- Each region maintains independent forecast accountability. Regional demand patterns, seasonal timing (summer in NA vs. EMEA), and SKU trends differ fundamentally, requiring separate regional forecasting.
- Key Metrics for regional heads to track - Forecast accuracy rate (>85%) and Margin %
- Cross-region sharing in exceptional cases basis COE team's approval

Channel Accountability Factors to Consider

BOPS (Buy Online Pickup Store)	Iron out clear policy and communication between channels. Data → decisions on which products to offer BOPS
Over-protection Penalties and Consumption Transparency:	Identified KPIs through regular dashboards: fill rate, # of shared pool requests, threshold breaches, sell-through.
Order Variance Alerts (>20%)	AI team → automated alerts/forecasts. Proactive alerting systems (data side)

Conclusion: KPIs, data, automation → company-wide operational efficiency.

Product Type Segmentation

Product Type	What It Is	Demand Pattern	Risk Level	Inventory Approach (example)	Sharing Rules
Evergreen	Core, replenishment basics	Predictable, stable	Low	SUP 30%	Least strict of the 3
Regular Seasonal	Standard seasonal fashion	Medium volatility	Medium	SUP 50%	Increase protections for volatility
Limited Editions / Designer/ Premium	Hype-driven, scarce SKUs	High volatility	High	SUP 70%	Most strict of the 3, focused on DTC satisfaction
Size Layer (All Types)	XS/XL slow; S/M/L normal	Varies	Medium	SUP 50%	Non-extreme sizes have more priority and restriction (size-level protection)

First Layer: Size Segmentation -> **Second Layer:** Limited/ Regular Seasonal -> **Residual SKUs**: Evergreen

As we move from low risk to high risk level SKUs, our supply protection increases (we would want to limit the shared pool %. For eg - for Evergreen category, we will restrict 30% of the demand to different channels and the rest goes to the shared pool.

Data Management & Governance

Data & Systems

- Shared inventory requires consistent SKU identifiers while allowing channel-specific tags to be applied at the fulfillment node.
- RFID or computer vision improves store inventory accuracy for omni fulfillment.
- Real-time data flow between systems prevents overselling and double allocation.
- Seasonal surges and one-off events require manual override capabilities without permanently altering system rules.

COE Governance

The COE is a cross-functional governance body that sets inventory rules, approves exceptions, and ensures shared inventory decisions protect margin, brand, and channel fairness.

- Sets protection limits by channel/product
- Approves exceptions (launches, spikes, delays)
- Resolves channel conflicts
- Monitors shared pool usage
- Reviews misallocation post-mortems

Other Valuable KPIs to Look Into

1. Shopping (Browser-to-Store-Visit Rate)
2. Checkout (Cart Abandonment Rate)
3. Fulfillment (OTIF)
4. Service (Channel Switch Rate)

Fulfillment Routing Logic (Learnings from Zara)

Teams need clarity on “where orders ship from” when multiple locations qualify (stores, warehouses, etc.)

Centralized Optimization Model

- **Issue:** Managing inventory transfers across 1,720 stores creates 2.9 million possible routes, making manual optimization infeasible.
- **Solution:** an optimization model that calculates the most profitable transfer by maximizing expected profit while considering demand, logistics, and business rules.

Built-In Business Rules

- **Issue:** Transfers must align with merchandising strategies, minimize store disruption, and adapt to different product lifecycle stages (e.g., redistribution vs. consolidation).
- **Solution:** Hardcoded constraints such as minimum display quantities, size distribution balancing, coverage thresholds, and limits on transfer quantities and destinations.

Demand-Responsive Forecasting System

- **Issue:** Retail demand is “short, uncertain, highly varying,” requiring rapid response to sales trends.
- **Solution:** A demand forecast model updated with recent weekly sales data, using linear regression and store-grouping techniques to refine predictions.

Returns & Reverse Logistics